

Get Real about HOW NAVAL AVIATION GOT BETTER

Safety doesn't come from magic bullets,
but from organizational learning.

BY JEFFERSON D. GRUBB

The first chapter of Tom Wolfe's 1979 work of narrative nonfiction, *The Right Stuff*, paints a lurid portrait of naval aviation in the 1950s.¹ Wolfe follows aviator and future astronaut Pete Conrad from funeral to funeral for squadron mates who had "augured in," been "burned beyond recognition," or otherwise "bought it" in aircraft mishaps. Conrad's experience was not unusual. In fiscal year 1953, the U.S. Navy and Marine Corps lost 723 aircraft and 423 people to aviation mishaps.² In contrast, in FY 2025, naval aviation lost just 11 aircraft and two people to aviation mishaps. Clearly, naval aviation got better. But how?

Naval aviation safety presenters usually display a graph that plots historic improvements in the annual rate of major mishaps by fiscal year since 1950, though the shape of the curve is consistent back to the beginning of such recordkeeping in 1922.³ (See figure 1.) The graph's steep initial decline tapers to a low, seemingly flat trend, a distinctive shape that gives the graphs their colloquial description—ski slope charts. In most cases, presenters annotate the slopes with text and arrows that indicate the timing of milestone changes to process and equipment. The annotations imply those things specifically called out are what drove down the mishap rate. That is, the charts imply naval



A F6F-3 Hellcat burns on board the aircraft carrier USS *Enterprise* (CV-6) on 10 November 1943. A sharp drop in the mishap rate near the start of World War II is likely explained by a decrease in reporting rather than a decrease in mishaps.



aviation got better through a succession of silver bullet fixes to various safety problems.

However, ski slope charts provide little actual evidence to support a causal relationship. In Aristotelian terms, they do not sufficiently differentiate between formal and efficient causes (see box on the facing page). In a survey of command briefs, books, journal articles, and web pages, the author examined 15 examples of annotated ski slope charts. Although each is annotated across its entire period, not all events are common to each. Especially post 1970, the charts' choices of highlighted events vary substantially. This suggests uncertainty about what changes significantly affected safety over the past half century.

Even when the same pre-1970 innovations are present, they often are located at different places on the curves. And, even when multiple charts list the same annotation at the same point, the curve rarely deflects downward there—improvement is not immediately associated with the timing. This raises doubts about whether or how any given annotated thing benefited safety. In other words, classic ski slope charts indicate exactly two things: that naval aviation got better and that naval aviation did stuff. However, the charts provide no evidence that the getting better was because of the particular stuff being done.

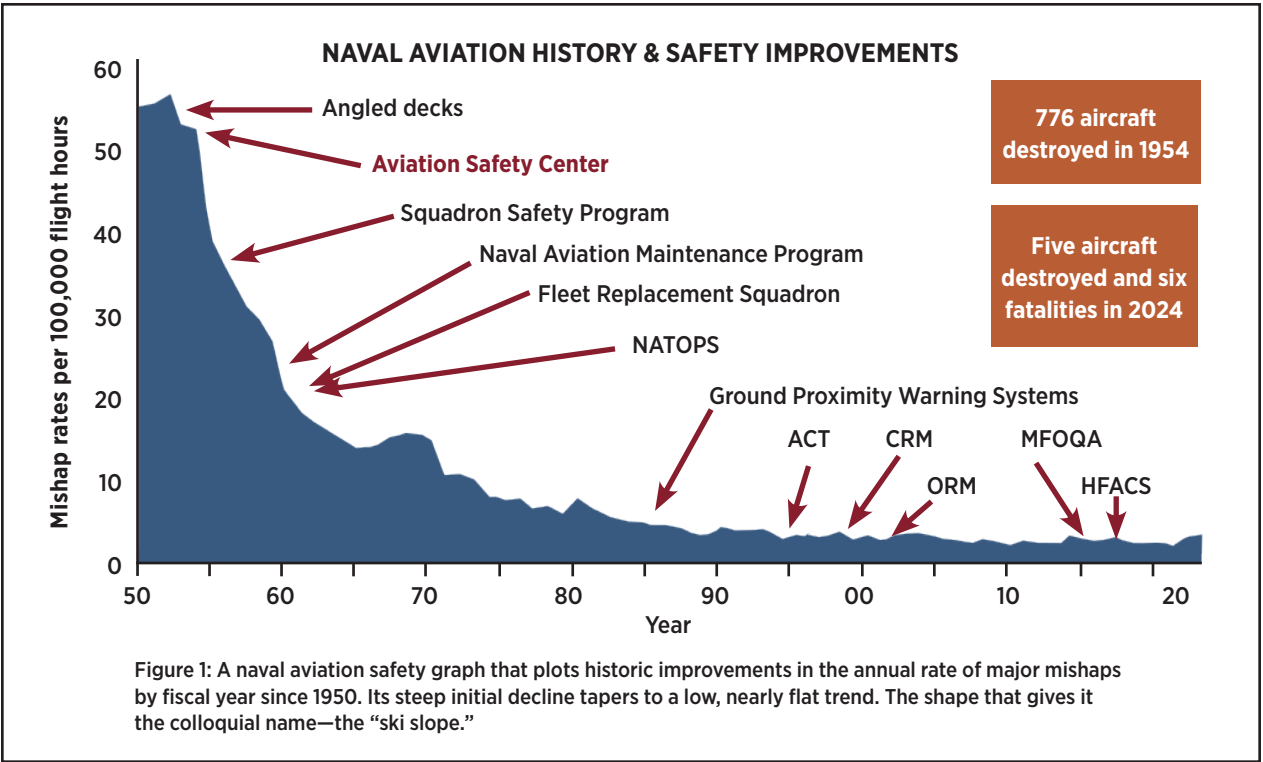
WHAT DON'T THE CURVES SAY

To better understand what drove the mishap rate down, look past the annotations and examine the curve itself.

From this larger perspective, both the steep initial decline and the recent flat trend of the curve are continuations of a single pattern that the mishap rate has followed over almost the entire history of U.S. naval aviation.

What is noisy data becomes a straight line when a formal analysis is performed, which indicates that the major mishap rate has the form of an exponential function. (The data analysis and methodology are discussed in the sidebar on page 26.) In social science terms, this exponential has the shape of what is commonly called a learning curve. Here, its downward (negative) shape is “eliminative.” That is, when smaller numbers indicate better performance (e.g., fewer errors, less time to complete a task, lower cost per item produced, etc.), task experience is reducing the measured quantity (accidents, for example).⁴ Business literature defines such organizational learning as changes in the organization's collective behavior that stem from experience.⁵ Improvement occurs through the gradual discovery, dissemination, and retention of best practices across the enterprise. In this light, the shape of the ski slope indicates naval aviation got better less by taking specific actions and more through some organizational learning process.

Although the overall learning curve by itself explains nearly all the variance in the major mishap rate, certain points in which the mishap rate changed significantly relative to its overall trend allow inferences to be drawn about what events significantly contributed to the learning, a process formally if somewhat obviously





NAVAL HISTORY AND HERITAGE COMMAND

ARISTOTLE SAYS that to explain why something exists or changes, four “causes” must be examined: material, formal, efficient, and final. For this article’s purposes, it is especially helpful to distinguish between two:

The *formal cause* of something—what Aristotle defines as the structure that makes something what it is. The blueprint, not the house.

And the *efficient cause*—for Aristotle, the agent who achieves the desired outcome. The carpenters.

In this case—loosely speaking—the formal cause can be thought of as the learning itself or becoming a learning organization that enables gradual discovery, dissemination, and retention of best practices. The efficient causes would roughly be the people, processes, and tools developed from and by that learning—the things the arrows on the ski slope chart point to and the people who conceived and carried them out, whether angled decks, safety officers, or NATOPS. And the end result—the house itself—is safety that is continually improving.

called “changepoint analysis.” Trivially, some of the key moments can readily be explained by probable changes in mishap reporting. Leaving those aside, the timing of other changepoints suggests a coherent story for how naval aviation got better.

ORGANIZATIONAL LEARNING

The annotations on classic ski slope charts emphasize many of the “discovery” elements of the effects chain. However, until a good idea is promulgated, it is merely a good idea; its dissemination takes time. For example, many ski slope charts have arrows pointing to 1951 with

The *Essex*-class USS *Antietam* (CVA/CVS-36), left. In 1953, the *Antietam* became the Navy’s first carrier with an angled deck, and was the only one until 1955. Twelve years later, the USS *Lake Champlain* (CVS-39) was the last carrier in service with only an axial flight deck.

the note “angled decks.” Although the U.S. and Royal Navies did indeed begin discussing the concept in 1951, angled decks could not affect the mishap rate until aviators began flying from them, which in the U.S. Navy began in January 1953 with test flights from its first permanent angled deck.⁶

For the next two years, that deck—belonging to the USS *Antietam* (CVA/CVS-36)—was the only angled one on the more than 20 active fleet carriers. Twelve years later, the *Essex*-class USS *Lake Champlain* (CVS-39) was still in service and still conducting fixed-wing operations from her axial flight deck, the last in the fleet.⁷ Consequently, the beneficial effects that angled decks had on the mishap rate could not have occurred prior to 1953, and the residual hazards of axial decks would still have affected the mishap rate through the *Lake Champlain*’s inactivation in late 1965. This is not to dispute that angled decks are safer, only to state that promulgating best practices and tools takes time.

The organization can influence how much time, however. The analysis revealed two consecutive periods of unusually rapid improvement in the mishap rate that together stretched from 1954 through 1965. As classic ski slope charts typically note, beyond angled decks, this period included a significant expansion of the forerunner to Naval Safety Command; the creation of aviation safety officer billets at all aviation activities; standardization of aviation maintenance practices; development of the Naval Aviation Training and Operations Standardization (NATOPS) program; and creation of RAGs—replacement air groups—squadrons that standardized the training for each type/model/series of aircraft (now known as fleet replacement squadrons).⁸

UNDERSTANDING THE MATH

ALTHOUGH THE COMMONLY USED ski slope charts usually present mishap rates since 1950, Naval Safety Command's annual major naval aviation mishap rate dataset stretches back to fiscal year 1922.

The mishap rate data indicate that naval aviation began classifying mishaps differently in 1936, so the short-term drop in both raw mishap numbers and year-to-year variability during the mid-1930s may be the result of an intentional change in how commands reported mishaps. Similarly, the sharp drop in the mishap rate at the beginning of the 1940s coincides with the onset of World War II. This period saw sudden, massive increases in operational tempo, aircraft performance, personnel accession, etc., making an actual drop in the mishap rate implausible. A more likely interpretation would be that, between competing priorities and alternative explanations for losses, the drop reflects a decrease in reporting rather than a decrease in mishaps.

Figure 2 plots the ski-slope mishap rate data from Figure 1 on a logarithmic scale. To do this, an analyst takes the y_1 values—the number of mishaps per 100,000 flight hours—and takes the natural logarithm of them: $\ln(y_1) = y_2$. That gets plotted on the same time-based (fiscal year) x axis. The resultant curve strongly resembles a straight line, and a regression

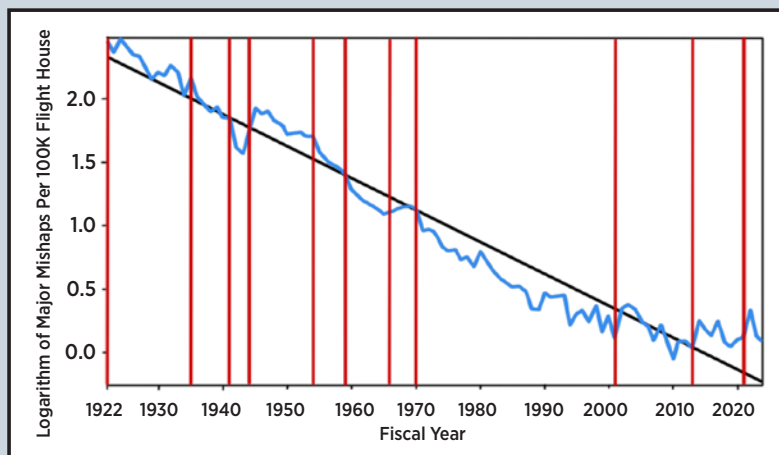


Figure 2: Changepoints are mapped with red lines against a logarithmic plot of annual major naval aviation mishap rates since 1922. The changepoints indicate the timing of significant changes in the mishap rate's behavior relative to its overall trend.

analysis produces a “best-fit” linear function, which indicates that the major mishap rate followed a negative exponential trend. In the behavioral sciences, negative exponential functions are characteristic of eliminative learning curves.

Although a formal regression analysis revealed that this negative exponential explains 95 percent of the variance in the major mishap rate from 1922 through 2025, Figure 2 also shows several extended periods during which the mishap rate remained consistently above or below the trend line. This suggests that certain events discretely raised or lowered the mishap rate relative to its overall trend.

To investigate this pattern, the author subtracted the overall trend

from the raw mishap rate data before performing a changepoint analysis on the residual data using the pruned exact linear time algorithm for mean and variance.¹⁰ This kind of analysis uses various tools, including binary segmentation and Bayesian estimation, to identify points at which the data's underlying pattern changes. These points can identify sudden, major shifts in structure—in this case, instances of localized changes to the mishap rate relative to its overall trend.

In Figure 2, these changepoints are marked with red vertical lines against the logarithmic plot of the raw data. This timing allows inferences to be drawn about events that significantly affected naval aviation safety—the efficient causes of improvements to or deterioration of safety.

Rather than mitigating specific hazards, these efforts all helped aggregate, disseminate, and ensure compliance with best practices, both particular and general. However, none of these sorts of adjustments guarantee lasting success; the organization can forget what it has learned. The changepoint analysis indicates mishap rates spiked during periods in which naval aviation had apparent incentives to disregard best practices. Although mishap reporting dropped with the onset of World War II, by war's end the reported mishap rate exceeded its prewar baseline by 18 percent. Similarly, the mishap rate jumped during the Vietnam War and with the onset of the war on

terror. Sustained combat operations have generally correlated with an increase in noncombat losses.

Although the military continuously plans and prepares for war, the country cannot continuously provide resources to cover every possible contingency. When conflict breaks out, the military has no choice but to meet the increased operational demands with the available resources. To surge personnel and matériel to the fight, leaders must accept the risk of deviating from established best practices. When this happens, mishap rates go up.

A resource-requirement mismatch may also explain the mishap rate increase after 2013. Under the



In this undated Vietnam War-era photo, a Douglas A-4 Shyhawk hangs in the catwalk of an aircraft carrier following a flight deck accident. The increase in noncombat mishaps during that period is likely explained by the tempo of sustained combat operations.

sequestration provisions of the Budget Control Act of 2011, the Department of Defense took a \$37 billion budget cut midway through fiscal year 2013.⁹ Prioritizing the Navy's operational requirements and fully funding other priority items resulted in cuts to maintenance and training budgets of about 30 percent, without commensurate cuts to operational requirements. Doing the same job with fewer resources necessarily entailed deviating from established practices. The changepoint analysis strongly suggests such deviations came at the cost of more mishaps.

AN ORGANIZATION THAT LEARNS OR A LEARNING ORGANIZATION?

If naval aviation got better via organizational learning, does that mean naval aviation is a learning organization? Not necessarily. Again, organizational learning refers to a process by which an organization's collective behavior changes with experience. All organizations learn. Although such learning can be purposeful and beneficial to the enterprise, it can also be haphazard and even

maladaptive. There even is a special term for maladaptive organizational learning: normalization of deviance.

In contrast, a learning organization is an organization that actively promotes beneficial organizational learning. Instead of asking whether naval aviation is one, it is more appropriate to ask *how much of one* it is. For aviation safety, the changepoint analysis shows that this varies over time. When naval aviation attends to collecting, disseminating, and enforcing best practices, it gets better faster. When it focuses on other things, learning slows or even reverses.

That brings us to the most recent changepoint and what to do about it. The naval aviation mishap rate again spiked relative to its historical trend in FY 2022. This jump does not neatly correlate with any obvious, single event, which has led to debate over the uptick's cause.

This debate is healthy, but the pattern of previous upticks suggests the quickest way to correct it is to ensure the enterprise is still doing those things it already knows work. Leaders need to honestly assess whether their organizations are still following established guidelines and procedures. If not, they must ask, Why not? Sometimes, there are good reasons for deviating from established practices, but doing so must be a conscious decision based on careful consideration of both the risks and potential rewards. If there is a good reason, it needs to be codified and shared so the entire organization can benefit. If not, we invite performance decay. Despite what the ski slope annotations suggest, silver bullet fixes are insufficient.

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